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KAMUS UMUM

constant NIL = NULL

constant IDX_UNDEF : integer = -1

type ElType : integer

type address : pointer to Node

type Node: < info: ElType;
 prev: address;
 next: address >

type List: < First: address,
 Last: address >

procedure insertFirst(input/output l : List, input x : ElType)

KAMUS LOKAL

new : Node

ALGORITMA

alokasi(new)

if(new $\neq$ NIL) then

new $\uparrow$.info $\leftarrow$ x

new $\uparrow$.next $\leftarrow$ NIL

if(l $\uparrow$.first = NIL and l $\uparrow$.last = NIL) then {list is empty}

l $\uparrow$.First $\leftarrow$ new

l $\uparrow$.Last $\leftarrow$ new

else

new $\uparrow$.next $\leftarrow$ l $\uparrow$.First

l $\uparrow$.First $\leftarrow$ new

procedure insertLast(input/output l : List, input x : ElType)

KAMUS LOKAL

new : Node

prev : address

ALGORITMA

if(l $\uparrow$.First = NIL and l $\uparrow$.Last = NIL) then

insertFirst(l, x)

else

alokasi(new)

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    if(new  $\neq$  NIL) then
        new $\uparrow$ .info  $\leftarrow$  x
        new $\uparrow$ .next  $\leftarrow$  NIL
    prev  $\leftarrow$  l $\uparrow$ .Last
    prev $\uparrow$ .next  $\leftarrow$  new
    l $\uparrow$ .Last  $\leftarrow$  new

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function search(l : List, x : ElType) $\rightarrow$ boolean

KAMUS LOKAL

found : boolean

p : address

ALGORITMA

found $\leftarrow$ false

if(l $\uparrow$.First $\neq$ l $\uparrow$.Last or l $\uparrow$.First $\neq$ NIL) then {l is not empty}

p $\leftarrow$ l $\uparrow$.First

while(p $\uparrow$.next $\neq$ l $\uparrow$.Last and p $\uparrow$.info = x) do

p $\leftarrow$ p $\uparrow$.next

if(p $\uparrow$.info = x) then

found $\leftarrow$ true

$\rightarrow$ found